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Late Precambrian (850–800 Ma) palaeomagnetic pole for the south Indian shield from the Harohalli alkaline dykes: geotectonic implications for Gondwana reconstructions

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Abstract

Palaeomagnetic results from nine sites of alkaline dykes in the Harohalli area, Dharwar craton, south India, provide the best palaeomagnetic pole for the Indian shield during 850–800 Ma. The mean palaeomagnetic direction is $D = 7.4°$; $I = 81.5°$; $n = 9$; $\kappa = 46$; $\alpha_{95} = 7.7°$) yielding a pole position at $\lambda = 28°S$; $\phi = 260°E$ (dp/dm = 14/15°). The results for the Harohalli dykes, together with earlier results, define a motion of the Indian subcontinent between ~1000 and 500 Ma and constrain an important geotectonic process between India and Africa during this period. While the 850–800 Ma Indian pole differs from the African pole for the same period in the Lotts and Rowley (1990) reconstruction, the Indian and African poles for 800–700 Ma and 650–500 Ma demonstrate excellent agreement. Thus, we suggest juxtaposition and collision between the African and the Indian continental nucleii along the Mozambique Belt (MB) to have occurred in the latest Precambrian (~800–750 Ma). Integration of our results with recent geochronological and petrological investigations in the then neighbouring Gondwana segments (India, Sri Lanka, Antarctica and Africa) helps to evaluate recently suggested geotectonic correlations between these fragments in the latest Precambrian. The south Indian granulite region extending up to the Bhavani shear zone could represent an independent Palaeoproterozoic terrane (SIGT) distinctive from the east coast region of India including the Eastern Ghat Mobile Belt (EGMB). The missing seafloor of the 'Mozambique Ocean' seems to have been consumed initially during the Late Precambrian (~800–750 Ma) by collision between Africa and the south Indian shield along the MB. The orogenic activity gradually migrated with time into the Highland/Southwestern Complex (HSWC) of Sri Lanka (~600–550 Ma) and then into the Lützow–Holm Complex (LHC) of Antarctica (~550–520 Ma) giving rise to final amalgamation of the continental blocks to form the Gondwana supercontinent.

1. Introduction

The Precambrian basement of the south Indian shield is an original Gondwana segment that became detached following Mesozoic–Cenozoic drifting. Reconstructions of the relative positions of Sri Lanka, India, Antarctica and Africa within their pre-drift configuration and the timing of their amalgamation into the supercontinent have been the subject of interest. Several recent studies have focused on demarcation and correlation of the petrological and age provinces within Sri Lanka, south India, Antarctica and Africa and have proposed Late Precambrian tectonic configurations (e.g., Katz, 1989; Kröner, 1991; Yoshida et al., 1992; Harris et al., 1994; Shiraishi et al., 1994). Palaeomagnetic data from these blocks are

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vital for such reconstructions, particularly from India because of its centrally located position within the reconstructions. However, until recently the Precambrian palaeomagnetic record from the Indian shield was scanty. The available data were not regarded of sufficient quality in terms of detailed demagnetisations, number of sites or lack of isotopic ages to enable meaningful comparisons between the continents. Our recent palaeomagnetic study on the mafic dykes from the granulite region of south India has proven important to overcome these defects, and has provided a good quality early-Meso- to Mesoproterozoic palaeomagnetic record for India. Here we present Neoproterozoic (850–800 Ma) palaeomagnetic results from alkaline dykes of the Harohalli area in south India.

The present study of the Harohalli alkaline dykes has great significance. Potassium–argon and whole rock and mineral isochron ages (Ikramudin and Stueber, 1976; Anilkumar et al., 1989; cf: Radhakrishna and Joseph, 1993) constrain an 850–800 Ma age with great certainty. Furthermore, no Late Precambrian palaeomagnetic pole has been available so far for the south Indian shield. Another simultaneous palaeomagnetic study on the Harohalli alkaline dykes by Dawson and Hargraves (1994) produced primary vectors for two dykes only, which is not adequate to compensate for the paleosecular variation of the geomagnetic field. Therefore, combination of our results with the data of Dawson and Hargraves (1994) enables determination of a high-quality palaeomagnetic pole of this age. The data have important tectonic implications for a Late Precambrian reconstruction of Gondwana.

2. Geological setting, petrography and age

The alkaline dykes of Harohalli are exposed south of Bangalore city between 12°38′ and 12°46′N and 77°23′ and 77°33′E. See Fig. 1 for a generalised geological setting and palaeomagnetic site locations. The terrain around Harohalli forms part of the Dharwar craton comprising a granitic complex with several N–S-trending greenstone belts. The local geology of the Harohalli area consists of a basement gneissic complex and the elongated body of the Closepet Granite, both traversed by various dyke intrusions. Available geochronological data constrain the ages of gneissic terrain and granite at $\geq$3.0 Ga and ~2.5–2.2 Ga, respectively (Naqvi and Rogers, 1987; Radhakrishna and Ramakrishnan, 1993). The dyke intrusions include metadolerites/norites, unmetamorphosed dolerites and alkaline dykes. Ikramuddin and Stueber (1976) have provided field and petrographic descriptions and whole-rock K–Ar and Rb–Sr dates of these dykes. The unmetamorphosed dolerites are olivine dolerite to quartz dolerite in composition and are predominantly oriented in NW–SE or E–W directions (Fig. 1). Seven whole-rock Rb–Sr isotope data yielded a regression line with an age of 2420 ± 246 Ma.

The alkaline dykes, whose palaeomagnetism is reported here, are classified petrologically as solvesbergite, tinguaite and bostonite and chemically as trachytes and phonolites (Ikramuddin and Stueber, 1976). Their trends vary between north (to NE) and south (to SW) with variable strike lengths (0.1 to 8 km) and widths (<1 to 15 m). At a few places, they cross-cut the dolerite dykes. The alkaline dykes are either green/grey or reddish brown in colour and consist almost entirely of alkali feldspar and alkali pyroxenes. The reddish brown dykes were described as altered dykes (Dawson and Hargraves, 1994). Diffusion of hematite staining and alteration of feldspars and cores of pyroxene grains is evident. The alteration is most likely deuteric. Ikramuddin and Stueber (1976) reported whole-rock Rb–Sr isotope data on eight samples yielding an isochron age of 832 ± 40 Ma and K–Ar whole-rock age of 810 ± 25 Ma for one of the dykes. Anilkumar et al., 1989 (cf. Radhakrishna and Mathew, 1993) reported an isochron age of 823 ± 15 Ma based on a combined Rb–Sr isotope data of whole-rock and alkali feldspar grains of the alkaline dykes. These results strongly attest a 850–800 Ma age for the alkaline dykes of Harohalli.

3. Palaeomagnetism

3.1. Methods

Oriented block samples were collected in the field using a magnetic compass. All samples were reoriented in the laboratory and 1-inch diameter and 1-inch long cylindrical specimens were drilled for palaeomagnetic measurements. Nine sites (dykes) were collected covering the total geographic area of

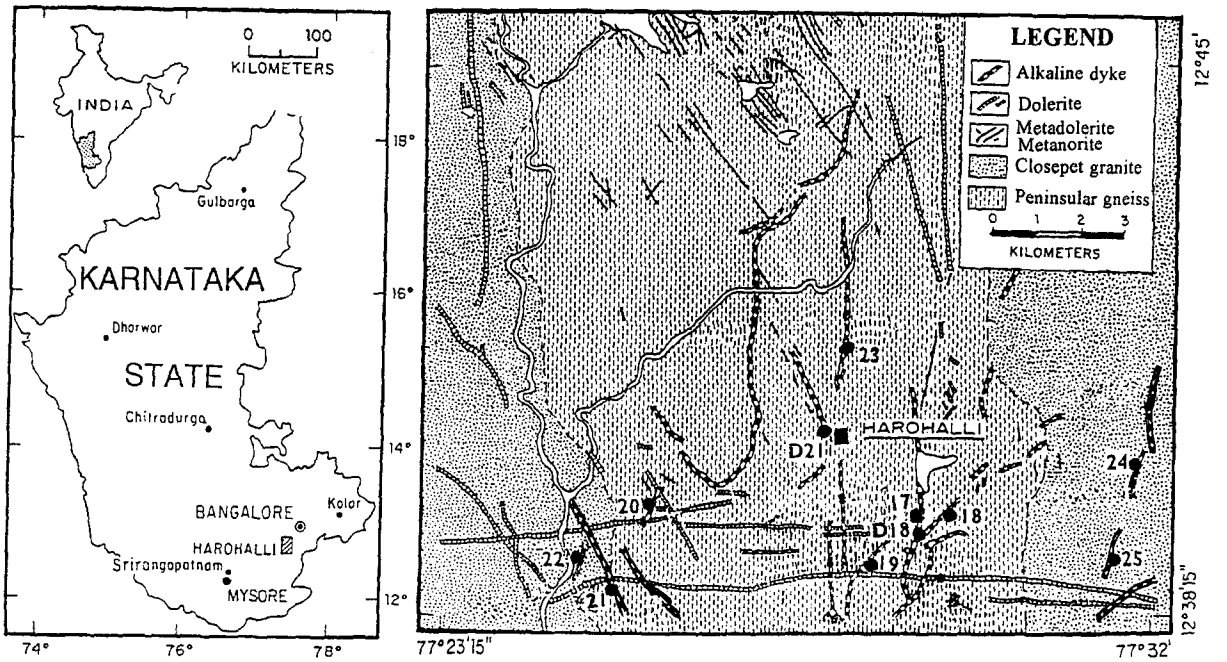


Fig. 1. Geological map of the Harohalli area showing the palaeomagnetic site locations. Numbers *17* to *25* are the sites of the present study and *D18* and *D21* are the sites of Dawson and Hargraves (1994).

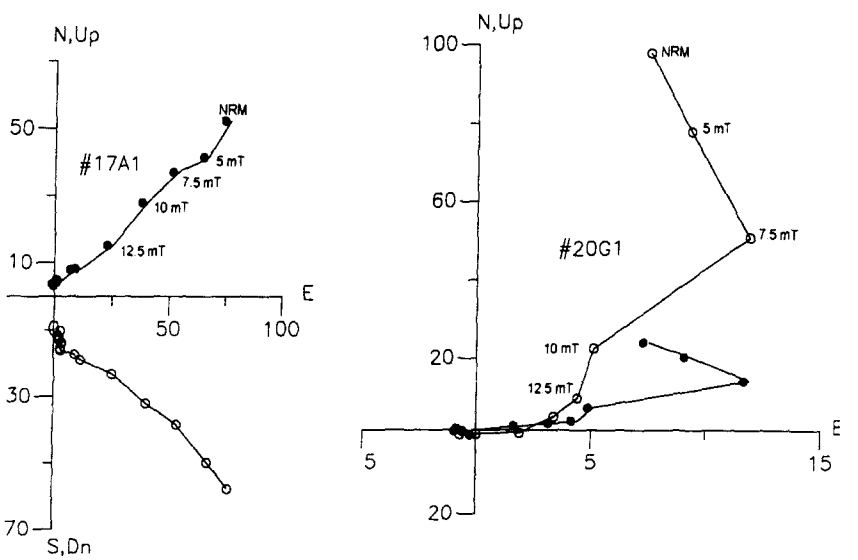


Fig. 2. Representative Zijderveld plots of AC demagnetisation of the Harohalli dykes. Scale in mA/m. Solid and open symbols represent magnetisation vector projected onto orthogonal horizontal and vertical planes, respectively.

the dyke exposures around Harohalli (Fig. 2). Six to eight block samples were taken from each site (dyke), thus a total number of 61 block samples were used in our palaeomagnetic study. Palaeomagnetic measurements were obtained with a MOLSPIN (UK) spinner magnetometer interfaced with a portable EPSON computer. The sensitivity of the instrument is of the order 0.5 mA/m. Progressive AF demagnetisations were performed on one specimen from each of the samples (i.e., total 61 specimens) at 5 mT and

thereafter at 2.5 mT steps until a stable magnetic component was isolated or till the intensities of magnetisation had dropped below the sensitivity of the instrument. A MOLSPIN AF shielded demagnetiser (UK), with a 100 mT peak field was used for AF demagnetisation. Thermal demagnetisations were performed on one specimen from each of the samples (except for site #23 whose samples were magnetically weaker than the instrument's sensitivity; i.e. in total 54 specimens) with progressive heating steps at 375, 420, 450, 480, 510, 530, 550, 560, 570, 580 and 590°C. Heatings were carried out with a MMTD-1 Thermal Demagnetiser (UK) capable of heating at a peak of 800°C. Demagnetisation data were analysed using Zijderveld vector plots, stereographic plots, and linear regression tests using principal component analysis (based on Kirschvink, 1980, modified by Halls, 1986). In the principal component analyses, only those linear trajectories of consecutive points with an angular uncertainty of $<10°$ were accepted. Individual site mean directions were calculated from each of the sample vectors and the mean swarm direction was obtained from the mean site directions following Fisher (1953) statistics.

3.2. Results

Samples from one of the nine sites (dykes, #23) proved almost devoid of magnetic minerals and the NRM remanence was too weak to be measured on the instrument. Samples of this site therefore were not considered for detailed AF or thermal demagnetisations. Samples from the remaining eight sites generally have low intensities ($\sim$1 A/m to 0.1 A/m) except for sites #20D (4 A/m) and #24F (20 A/m). Samples of these sites have scattered NRM directions with declinations due east or north and either positive or negative steep to shallow inclinations. AF demagnetisation at a 10–15 mT peak field generally proved useful to isolate stable directions in a majority of the samples (Fig. 2). The directions in many samples remained unchanged up to $\geq$570°C (Fig. 3). Specimens 17B, 19F and 20G showed some change upon thermal demagnetisation (Fig. 3) and the directions between 450 and 550°C closely compare with the directions of characteristic magnetisation achieved in the other samples of the sites upon AF demagnetisation. Although five out of six samples in site #18 gave stable directions, these vectors are highly scattered and no single consistent vector could be isolated. Stable directions from all samples of each of the other sites do not show perfect within-site clustering; however, the majority of the samples (60%) from these sites cluster at steep positive inclinations, whereas site #25 shows a reverse vector with steep negative inclination (Fig. 4). No favourable sampling sites could be located for baked contact tests. However, because all sites show good between-site grouping with a reverse vector in one of the dykes and because the site vectors do not coincide with any other known post-800 Ma direction from south India, the mean vector of the Harohalli alkali dykes ($D = 7.4°$; $I = +81.5°$; Table 1) is considered as the direction of primary magnetisation. While there is good within-site grouping (Fig. 4), there is also a pronounced within-site longitudinal spread and some of the samples show a reverse magnetisation (for example 20G, Figs. 2 and 3). This indicates different cooling histories for the various dykes and we take it that acquisition of remanence occurred over a protracted cooling period, thus the data would tend to average out the geomagnetic secular variation. The study of Dawson and Hargraves (1994) also isolated a similar characteristic magnetisation direction in the two sites studied, which closely match our results. All samples from their site #D21 dyke (different from our sites) were thermally demagnetised, and their site #18 samples (from our site #18 dyke) were thermally demagnetised up to 400°C followed by AF demagnetisation between 20 and 45 mT. Combination of these well agreeing directions of the two independent studies produces a good primary remanence result for the Harohalli alkaline dykes. The results of our palaeomagnetic data along with the data of Dawson and Hargraves (1994) are compiled in Table 1 and are depicted in Fig. 4 with α_{95} circles.

4. Discussion

4.1. Implications for the Neoproterozoic APWP of India and the Gondwana model

Our present report of palaeomagnetic data for the Harohalli alkaline dykes provides a high-quality pole averaging secular geomagnetic variation. The

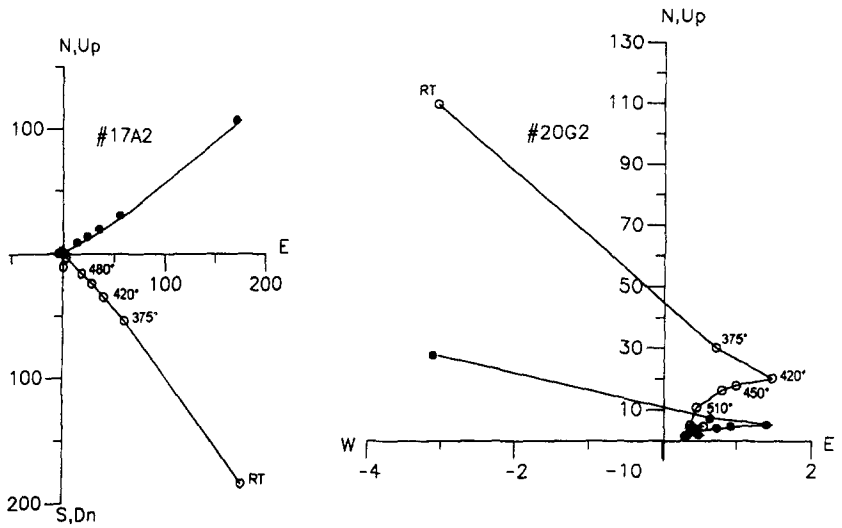


Fig. 3. Representative Zijderveld plots of thermal demagnetisation of the Harohalli dykes. Scale in mA/m. Symbols as in Fig. 2.

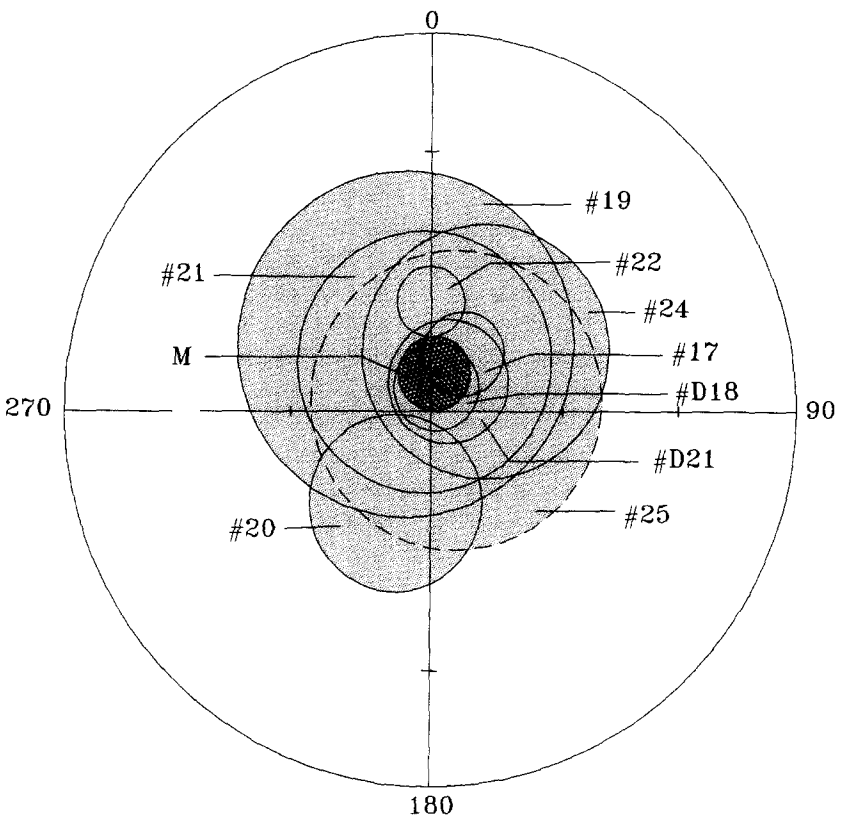


Fig. 4. Equal-area projections of mean palaeomagnetic directions of the Harohalli dykes as circles of confidence (at the 95% level). Continuous-line circles indicate positive inclinations and the discontinuous-line circle indicates negative inclinations. Numbers *17* to *25* are the sites of the present study and the data of two sites (*D18* and *D21*) of Dawson and Hargraves (1994) are also incorporated. *M* denotes the circle of the mean direction (with its centre marked as dot) of the combined data.

Table 1
Palaeomagnetic data of the Harohalli alkaline dykes

Site No.	Trend	Lat. °N	Long. °E	N	N_s	D	I	κ	α_{95}	λ	ϕ
#17[a]	N–S	12.66	77.49	8	8(5)	30.9	75.2	77	9	35.8S	274.7 E
#18	NE–SW	12.65	77.49	6	5	No grouping					
#19[a]	NE–SW	12.64	77.47	7	3(3)	337.9	73.9	11	38	39.9S	243.3 E
#20[a]	NE–SW	12.43	77.65	7	6(4)	200.5	68.5	43	19	7.8S	251.3 E
#21[a]	NW–SE	12.42	77.64	8	8(3)	355.1	79.1	18	29	33.4S	255.5 E
#22[a]	NNW–SSE	12.41	77.64	5	4(3)	0.5	65.7	239	8	54.5S	258.2 E
#23	N–S	12.68	77.46	7		Low intensities; no stable directions					
#24[a]	NNW–SSE	12.67	77.52	7	7(4)	43.7	71.5	12	28	35.1S	285.5 E
#25[a]	NNW–SSE	12.64	77.52	6	4(3)	67.4	−82.8	15	33	6.9S	244.4 E
D18[a]	N–S	12.66	77.49	6	5(5)	18.4	83.9	65	10	23.8S	261.6 E
D21[a]	NW–SE	12.68	77.46	6	6(6)	32.0	82.0	28	13	25.6S	266.8 E
Mean of 9 sites						7.4	81.5	46	7.7	28S	260 E
						dp/dm = 14/15°					

N = number of samples studied from the site; Ns = number of samples which yielded stable directions, and the number of samples which show grouping within brackets; D and I are declination and inclination (in degrees), respectively, of the characteristic remanent magnetic vectors; κ = precision parameter; α_{95} = radius of the 95% confidence cone about the mean direction; λ and ϕ are respectively the latitude and longitude of VGP.

[a] sites included to calculate mean swarm direction. D18 and D21 are the data sites of Dawson and Hargraves (1994).

available isotope ages for these dykes indicate an 850–800 Ma age for the palaeomagnetic pole. No other palaeomagnetic data for this period exist for the Indian subcontinent. Thus the pole is potentially an important contribution to the south Indian shield palaeomagnetic data base. Apart from the Harohalli dykes, isotopic age control is available for a palaeomagnetic result for the Malani rhyolites (745 ± 10 Ma; Klootwijk, 1975). Hargraves and Bhalla (1983) assigned an age of 980 ± 110 Ma to the directions of a few dykes, metashale and sandstone in the Cuddapah Basin. The palaeomagnetic directions of these rocks are also comparable to the directions of Wajrakarur Kimberlite (cf. Radhakrishna and Joseph, 1993); the Kimberlite has been dated at 1090 Ma, based on concordant Rb–Sr isotope data of phlogopite megacrysts from four pipes (Anilkumar et al., 1993). Therefore, the combined data of these rocks may define an improved Apparent Polar Wander Path (APWP) for the Indian shield for the latest Proterozoic since 1000 Ma.

The Neoproterozoic–early Palaeozoic Indian pole data have been rotated into the African coordinates following Lottes and Rowley (1990) reconstruction and plotted in comparison with the African pole data in Fig. 5. It is evident that the Indian APWP shows a large shift for the 1000–500 Ma interval, from high latitudes around 1000 Ma to equatorial latitudes at the time of the Harohalli dyke emplacement, and then a significant longitudinal shift up to the early Palaeozoic. These displacements correspond to drift rates of more than 0.6°/yr.

The palaeomagnetic record for the Neoproterozoic to early Phanerozoic of India has important implications for the Gondwana configuration, especially between the African and the Indian segments. McWilliams (1981) proposed a Gondwana reconstruction for this period on the basis of palaeomagnetic data. In absence of Indian data, other than for the ~750 Ma Malani rhyolites, his analysis was largely centred on a discussion of apparent polar motions between Africa and Australia. According to McWilliams (1981), the east and west Gondwana segments were, in the Neoproterozoic, quite different and independent crustal blocks and merged in the mid-Late Cambrian following collision tectonics in the latest Precambrian–early Palaeozoic. Our present analysis of pole data (Fig. 5) suggests that India was separate from the east African (Congo and Kalhara) cratons during the emplacement of the Harohalli dykes (850–800 Ma), but appear juxtaposed by 800–750 Ma. This deduction supports the broad inference of collision between the east and west Gondwana segments, but would place the time of collision at

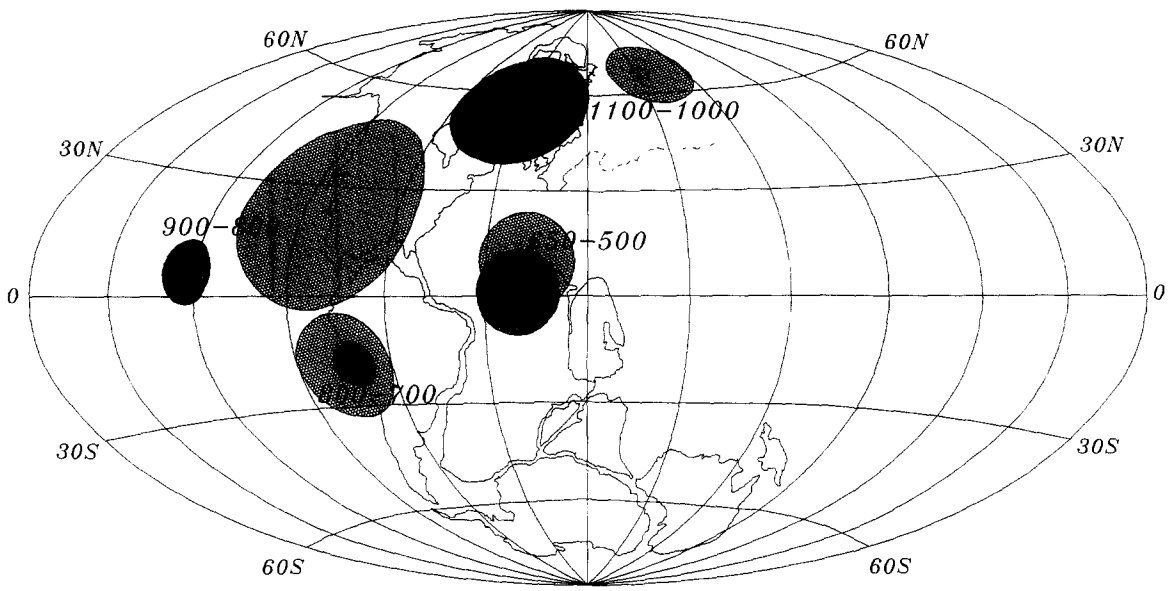


Fig. 5. Mean palaeomagnetic pole data in the form of circles of confidence (95% level) for India (dark shade) and Africa (light shade) during the late Precambrian to early Palaeozoic. The Indian data are rotated into the African coordinates using the rotation parameters of Lottes and Rowley (1990) for reconstruction of continents shown in the figure. The data used are detailed in the Appendix.

about 800 Ma, i.e., considerably earlier than the period proposed by McWilliams (1981). Our argument is in agreement with the suggested ~800 Ma collision based on isotope dates of syntectonic granites in the Mozambique belt (Pinna et al., 1993).

4.2. Implications for geological correlations and reconstructions of Gondwana

Regional correlations and configurations of crustal blocks of the eastern Gondwana have been reported recently based on correlations of metamorphic facies and isotopic dating. The present palaeomagnetic results of the Harohalli alkaline dykes permit us to evaluate and further constrain such reconstructions. The review of McWilliams (1981) is taken as a starting point for such discussion. He proposed that the N–S-trending Mozambique Belt (MB) in Africa was the leading edge of the collision process and a cryptic suture could be lying in western coastal India, Antarctica and/or possibly in some of the segmented plateau in the Indian Ocean. Cordani and Cooray (1989) and Kröner (1991) identified the tectonic events of the MB in the Sri Lankan geology based on lithostratigraphic correlations. Sri Lanka consists of three lithostratigraphic units: the westerly Wani Complex (WC), the central Highland/SouthWestern Complex (HSWC) and the easterly Vijayan Complex (VC) (see Fig. 6); the WC and HSWC had experienced major structural and metamorphic development during 950–550 Ma and represent the Sri Lankan extension of the MB. In India, the Kerala Khondalite Belt (KKB), to the south of the Achankovil shear zone (AC, in Fig. 6) was regarded as a possible extension of the MB–WC–HSWC belt (e.g. Shiraishi et al., 1994) based on a few ages of incipient charnockitisation (Srikantappa et al., 1985; Hansen et al., 1985; Choudhary et al., 1992) and remote sensing data (Drury et al., 1984). However, Radhakrishna et al. (1990) pointed out that the AC does not constitute a major shear zone demarcating the boundary between two different crustal blocks because of structural continuity across the 'shear' zone. Harris et al. (1994) have supported this conclusion because of the absence of a distinct crustal history in the Nd T_{DM} ages on either side of the AC.

Harris et al. (1994) proposed that the South Indian Granulite Terrain (SIGT) in India, the WC and HSWC in Sri Lanka and the Rayner Complex (RC) and the Lützow–Holm Complex (LHC) in Antarctica constitute a distinct tectonic block that is juxtaposed

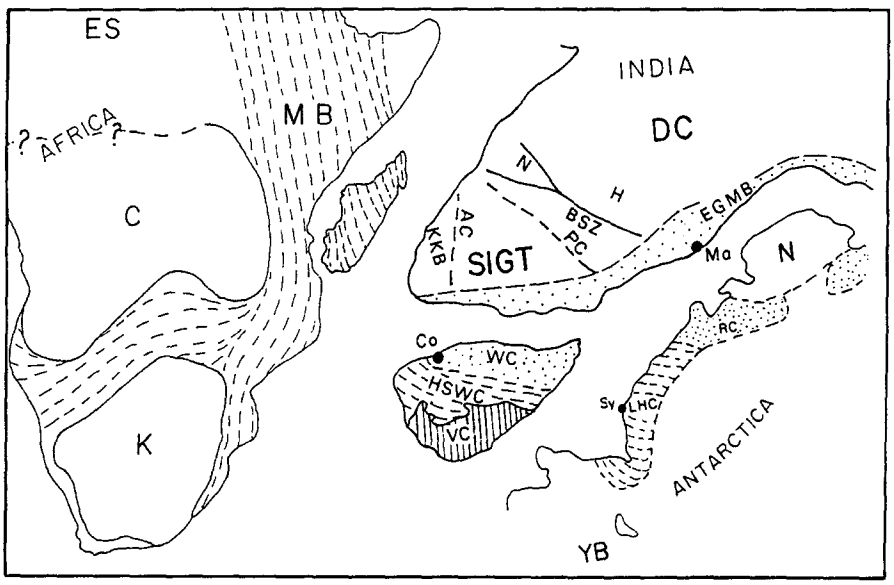


Fig. 6. Gondwana reconstruction showing tectonic correlations between Africa, India, Sri Lanka and East Antarctica. *Co* = Colombo; *Ma* = Madras; *Sy* = Syowa station; *LHC* = Lutzou Holm Complex; *RC* = Rayner Complex; *N* = Napier Complex; *YB* = Yamato–Belgica Complex; *VC* = Vijayan Complex; *HSWC* = Highland/Southwest Complex; *WC* = Wani Complex; *K* = Kalahari craton; *C* = Congo Craton; *ES* = East Sahara; *MB* = Mozambique Belt; *SIGT* = South Indian Granulite Terrane; *EGMB* = Eastern Ghat Mobile Belt; *DC* = Dharwar Craton; *KKB* = Kerala Khondalite Belt (Trivandrum Block according to some authors); *N* = Nilgiri Massif; *BSZ* = Bhavani shear zone; *AC* = Achankovil 'shear' zone; *PC* = Palghat Cauvery 'shear' zone; *H* = Harohalli. Details in text.

to the northern Dharwar craton along the Palghat Cauvery Shear Zone (PC in Fig. 6) in India. According to them, their Nd T_{DM} ages from the south Indian terrain and those reported from the WC and HSWC in Sri Lanka (Milisenda et al., 1988) fall between 3.0 and 1.1 Ga in contrast to the T_{DM} ages (1.4–1.1 Ga) in the VC. However, the original work of Milisenda et al. (1988) documented quite distinct Nd T_{DM} ages for the three lithotectonically distinct units; the HSWC had T_{DM} ages >2.2 Ga while the WC and VC had T_{DM} ages of only 1–2 Ga. Hiroi et al. (1994) also revealed contrasting prograde *P–T* evolution for the HSWC and the WC. Shiraishi et al. (1994) superposed U–Pb zircon ages and reinforced the Milsenda et al. (1988) distinction of T_{DM} ages among the WC, HSWC and VC without exception; high-grade metamorphism had occurred during 610–550 Ma in the HSWC and ~1100 Ma in the WC. Therefore, we suggest that the SIGT with a late Archaean–Palaeoproterozoic crustal history (mean Nd T_{DM} ages of 2.4 ± 0.3 Ga; except for sample #19 of their study) might represent an independent crustal block. In the absence of convincing evidences for crustal welding along the PC, we suggest that the SIGT extends beyond the PC to the north, upto the Bhavani Shear Zone (BSZ), along which it might have been juxtaposed to the Dharwar craton (DC). The BSZ exhibits undisputed shear features with substantial displacements of crustal blocks. Such a process can explain the similarity of T_{DM} ages to the south and north of the PC, upto the BSZ. The Palaeo- to Mesoproterozoic palaeomagnetic directions of the dykes in the SIGT are closely similar to the dyke directions reported from the DC (Radhakrishna and Mathew, 1993). Although good age data are presently lacking for the directions from the DC, such close similarity, in the absence of recognisable younger geological events, has been interpreted to indicate that the SIGT and the DC have been together as a combined south Indian shield at least since the Palaeoproterozoic (~2000 Ma; ibid). The K–Ar and Rb–Sr isotope ages (~2000 Ma) of unsheared/unmetamorphosed dykes (Radhakrishna et al., 1995) and Rb–Sr biotite ages in the western BSZ (Peucat et al., 1993) also support welding of these blocks in the Palaeoproterozoic. Therefore, the present palaeomagnetic pole of the Harohalli alkaline dykes and its interpretations in the preceding

section would apply to the combined south Indian shield.

Shiraishi et al. (1994) correlated the HSWC of Sri Lanka with the LHC of Antarctica. They suggested that the HSWC and the LHC were continuous orogenic belts, intensely deformed and having experienced granulite facies metamorphism at ~610–550 Ma and 550–520 Ma, respectively. These correlations, in combination with the correlations between the MB in Africa and the HSWC in Sri Lanka, suggest that the MB orogenic belt developed consequent to the ~800 Ma collision between the south Indian shield and African continental nucleii (with metamorphic ages of 750–650 Ma); the orogenic belt further extended through the HSWC in Sri Lanka into the LHC in Antarctica (Fig. 6). It is interesting to find that the ages of orogeny become younger from Africa, through Sri Lanka, into Antarctica segments. Similar progressively younger orogeny and collision events along the Indus suture from west to east in the Himalayan region were interpreted to reflect collision to have progressed from western to eastern regions (for example, Radhakrishna, 1987). Applying the same analogy, the orogenic activity initiated by the ~800 Ma collision between Africa and India along the MB is proposed to have gradually migrated with time into Sri Lanka (HSWC) and then to Antarctica (LHC) segments leading to the final amalgamation of Gondwana segments into a supercontinent.

Acknowledgements

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Appendix A

Neoproterozoic poles used for comparisons between India and African crustal blocks. Indian poles rotated to African coordinates using the Lottes and Rowley (1990) reconstruction. Indian data are taken from the compilations of Klootwijk (1979), Hargraves and Bhalla (1983), Radhakrishna and Joseph (1993) and Vander Voo (1993). African data are compiled from Brock (1987), Wilson et al. (1987), Allsopp et al. (1989), Briden et al. (1993) and Vander Voo (1993).

	λ	ϕ	α_{95}	Age (Ma)
1100–1000 Ma				
India:				
Tirupati dykes	−4.8	335.9		~1000
Metashale, Cuddapah Basin	−22.6	336.1		~1000
Metasandstone, Cuddapah	−22.3	338.8		~1000
Visakhapatnam Charnockite	14.0	9.0		~1000
Tiptur dyke	13.6	31.1		
Tiruvannamalai dyke (is an overprint)	6.5	347.8		
Wajrakarur Kimberlite	18.3	6.1		~1090
Mean:	0.5	346.0	17	
Mean rotated to Africa	55	329		
Africa:				
Umkonda lavas and dykes	65.0	40.0		~1080
Barby Formation	68.0	20.0		~1000
Post Waterberg diabases	65.0	50.0		~1115
Mean:	66.5	37.4	10	
900–800 Ma				
India:				
Harohalli alkaline dykes	−28.0	260.0	8	850–800
Rotated to Africa	6	237		
Africa:				
Klein Karas dykes	20.0	294.0		900–850
Ikorongo Group	35.0	264.0		
Kigonero Flags	12.0	273.0		
Mbala dolerites	−9.0	280.0		900–800
Mean:	22.8	277.5	28	
800–700 Ma				
India:				
Malani rhyolites	−79.3	135.0	6	~750
Rotated to Africa	−18	288		
Africa:				
Gagwa lavas	−29.0	283.0		~800
Bukoban Intrusives	−11.0	281.0		~800
Manayovu red beds	−24.0	298.0		
Mean:	−18.4	285.0	14	

	λ	φ	α_{95}	Age (Ma)
650–500 Ma				
India:				
Bhandar sandstone (mean of 3)	−44.4	30.2		~600–500
Bhagawanwala sandstone	−22.1	31.8		~Mid-Camb.
Khewra sandstone	−28.4	31.0		~570–540
Rewa sandstone (mean of 2)	−41.4	31.0		~Latest Proterozoic
Mean:	−34.0	31.0	12	
Mean rotated to Africa	1.2	340		
Africa:				
Ntonya Ring structure	18.0	337.0		~520
Hook Intrusives, Zambia	14.0	336.0		~500
Sijarira Group, Zambia	4.0	351.0		~620–540
Nama Group (N3)	−2.0	344.0		~580–480
Mean:	9.8	341.0	14	

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